

B.1 Thermal energy transfers

Topic warm-up · IB Physics (2023 syllabus)

with worked answers

Name: _____

Date: _____

Total: **27 marks**

$c_{\text{water}} = 4200 \text{ J kg}^{-1}\text{K}^{-1}$; $L_{\text{f, ice}} = 3.34 \times 10^5 \text{ J kg}^{-1}$. Show your working.

Question 1 — The marshmallow debrief

(5 marks)

Around a campfire, three friends argue. One says the fire “contains a lot of heat”; another says the marshmallow’s temperature “is the energy in it”.

- (a) Define temperature in terms of the behaviour of molecules. [2]
- (b) Define the internal energy of a substance, and state how it differs from temperature. [2]
- (c) Correct the first friend: state what heat actually is. [1]

Question 2 — The hot chocolate factory

(6 marks)

A café’s 1.8 kW steam wand heats 0.80 kg of water from 18 °C to 88 °C for hot chocolate. Assume no losses.

- (a) Calculate the energy required. [2]
- (b) Calculate the heating time, in seconds. [2]
- (c) A barista doing this calculation got 0.13 s. Identify the unit error that produces an answer exactly 1000 times too small. [2]

Question 3 — The ice bath challenge

(9 marks)

After training, a rower tips 0.30 kg of ice at 0 °C into an insulated tub holding 2.0 kg of water at 25 °C.

- (a) State why the temperature of the ice–water mixture stays at 0 °C *while* ice is melting, even though energy is flowing in. [2]
- (b) Calculate the energy needed to melt all the ice. [2]
- (c) Show that the final temperature of the water is about 11 °C. [4]
- (d) A teammate’s attempt gave about 22 °C because one term was left out. Identify the missing term. [1]

Question 4 — The backyard pizza oven

(7 marks)

A spherical pizza oven radiates approximately as a black body.

- (a) State Stefan–Boltzmann’s law, defining each symbol. [2]
- (b) The oven’s absolute temperature is doubled during preheating. Determine the factor by which its radiated power increases. [2]
- (c) A second oven has twice the radius *and* twice the absolute temperature. Determine the factor by which its radiated power exceeds the original’s. [3]

Answers

Question 1 — The marshmallow debrief

- (a) A measure of the average random kinetic energy per molecule.
- (b) The total of the random kinetic energies *and* the intermolecular potential energies of all the molecules. Temperature is an average *per molecule*; internal energy is a *total* for the substance, so it also depends on how much substance there is.
- (c) Heat is energy *in transfer* between bodies because of a temperature difference. A body does not “contain” heat; it contains internal energy.

Question 2 — The hot chocolate factory

- (a) $Q = mc\Delta T = 0.80 \times 4200 \times 70 = 2.35 \times 10^5 \text{ J}$ (235 kJ).
- (b) $t = Q/P = 235\,200/1800 = 130 \text{ s}$ (about 2.2 minutes).
- (c) The energy was left in kilojoules while the power was in watts: $235.2/1800 = 0.13 \text{ s}$. Mixing the kilo prefix into one side of the division only, i.e. forgetting that $235 \text{ kJ} = 2.35 \times 10^5 \text{ J}$. (Misreading 1.8 kW as 1.8 W is the opposite slip: it makes the time 1000 times too *large*.)

Question 3 — The ice bath challenge

- (a) While ice is melting, the incoming energy goes into breaking the intermolecular bonds of the solid (latent heat), not into raising the average molecular kinetic energy. Temperature tracks the latter, so it does not change.
- (b) $Q = mL_f = 0.30 \times 3.34 \times 10^5 = 1.0 \times 10^5 \text{ J}$.
- (c) The warm water supplies both the melting *and* the warming of the resulting meltwater from 0°C to T :

$$2.0(4200)(25 - T) = 1.002 \times 10^5 + 0.30(4200)T$$

$$210\,000 - 8400T = 100\,200 + 1260T \Rightarrow 109\,800 = 9660T$$

so $T = 11.4^\circ\text{C} \approx 11^\circ\text{C}$.

- (d) The latent heat term, $mL_f = 1.0 \times 10^5 \text{ J}$. Leaving it out treats the ice as though it were simply 0.30 kg of water already at 0°C , which gives $T = 210\,000/9660 = 21.7 \approx 22^\circ\text{C}$: far too warm, because melting the ice is by far the largest energy cost in the problem.

Question 4 — The backyard pizza oven

- (a) $P = \sigma AT^4$ (taking emissivity 1 for a black body), where σ is the Stefan–Boltzmann constant, A the surface area and T the *absolute* temperature.
- (b) $2^4 = 16$.
- (c) Area scales as r^2 , giving a factor $2^2 = 4$; temperature contributes $2^4 = 16$. Together $4 \times 16 = 64$.